ACS Module-2.

$x' = Ax + B$
$y = Cx + D$.

(Q) Find state transfer matrix TF.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \overset{A}{\begin{bmatrix} 0 & 3 \\ -2 & -5 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \overset{B}{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}} u$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \overset{C}{\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Formula: $\boxed{T.F = C[SI - A]^{-1} B + D}$

Ans) $$[SI - A] = \begin{bmatrix} S & 0 \\ 0 & S \end{bmatrix} - \begin{bmatrix} 0 & 3 \\ -2 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} S & -3 \\ +2 & S+5 \end{bmatrix}$$

$$[SI - A]^{-1} = \frac{\begin{bmatrix} S+5 & 3 \\ -2 & S \end{bmatrix}}{S(S+5) - (-3 \times 2)}$$

$$= \frac{\begin{bmatrix} S+5 & 3 \\ -2 & S \end{bmatrix}}{S(S+5) + 6}$$

$$= \frac{\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} s+5 & 3 \\ -2 & s \end{bmatrix}}{s(s+5)+6}$$

$$= \frac{\cancel{\begin{bmatrix} 2s+10 & 3 \\ -2 & 0 \end{bmatrix}}}{\cancel{s^2+5s+6}}$$

$$= \begin{bmatrix} \cancel{(2s+10-2)} & \cancel{(2\times3+5)} \end{bmatrix}$$

$$= \frac{\begin{bmatrix} (2\times(s+5) + 1\times -2) \; \cancel{(2s+10-2)} & 2\times(\overset{3}{\cancel{6+s}}) + 1\times s \\ 1(s+5) & (1\times 3). \end{bmatrix}}{s^2+5s+6.}$$

$$= \frac{\begin{bmatrix} 2s+8 & 6+s \\ s+5 & 3 \end{bmatrix}}{s^2+5s+6.}$$

$2s+8 \quad 6+s$

$2s+8-6-s$

$s+2$

$C(sI-A)^{-1}B+D$

$$= \frac{\begin{bmatrix} \frac{2s+8}{s+5} & \frac{6+s}{3} \end{bmatrix}}{s^2+5s+6} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 0$$

$$= \begin{bmatrix} \frac{3s+14}{s^2+5s+6} & \frac{3s+14}{s^2+5s+6} \\ \frac{s+8}{s^2+5s+6} & \frac{s+8}{s^2+5s+6} \end{bmatrix}$$

(Q) Find Transfer matrix TF.

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

$\dot{x} = Ax + Bu$.

$y = Cx + Du$.

n) $T\cdot F = C\left[SI - A\right]^{-1} B + D$

$$SI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 \\ 6 & +5+s \end{bmatrix}$$

$s(s+5) - (-1\times 6) = s^2+5s+6$

$$(SI - A)^{-1} = \frac{\begin{bmatrix} s+5 & 1 \\ -6 & s \end{bmatrix}}{s^2+5s+6}$$

$$C(SI-A)^{-1} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s+5 & 1 \\ -6 & s \end{bmatrix}$$

R C R C

$1 \times 2 \quad 2 \times 2$

$$= \frac{\begin{bmatrix} s+5 & 1 \end{bmatrix}}{s^2+5s+6}$$

$1 \times 2 \quad 2 \times 1$

$C(SI-A)^{-1}B$

$$= \frac{\begin{bmatrix} s+5 & 1 \end{bmatrix}}{s^2+5s+6} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{\begin{bmatrix} s+5 & \cancel{0} \\ \cancel{0} & \cancel{0} \end{bmatrix}}{s^2 + 5s + 6}$$

$$= \frac{s+5}{(s+3)(s+2)}$$

(Q) For the system

$$\dot{x} = \overset{A}{\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}} x + \overset{B}{\begin{bmatrix} 1 \\ 1 \end{bmatrix}} u$$

$$y = \overset{C}{\begin{bmatrix} 4 & 0 \end{bmatrix}} x$$

given $x_1(0) = 0 \quad x_2(0) = 1$

Find

(a) state transition matrix $\phi(t)$.

(b) time response.

(c) Solution of state equation if input is unit impulse.

(d) Also output when input is unit input.

Standard values:

$$L^{-1}\left[\frac{1}{s-a}\right] = e^{at}$$

$$L^{-1}\left[1/s\right] = 1$$

$$L^{-1}\left[\frac{1}{s^2+a^2}\right] = \frac{\sin at}{a}$$

$$L^{-1}\ s/s^2+a^2 = \cos at$$

$$L^{-1}\left[\frac{1}{[s+a]^n}\right] = e^{-at}\,\frac{t^{n-1}}{(n-1)!}$$

unit step impulse

$$\delta(t) = u(s) = 1$$

unit step input

$$u(t) = u(s) = 1/s$$

unit ramp input

$$r(t) = u(s) = \frac{1}{s^2}.$$

(a) $\boxed{\phi(t) = L^{-1}[sI - A]^{-1}}$

$$[sI - A] = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

$$[sI - A] = \begin{bmatrix} s-2 & 0 \\ 0 & s-4 \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} s-4 & 0 \\ 0 & s-2 \end{bmatrix} \Big/ (s-2)(s-4)$$

$$L^{-1}\left[(sI-A)^{-1}\right]$$

$$= L^{-1}\begin{bmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-4} \end{bmatrix}$$

$$\phi(t) = \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{4t} \end{bmatrix}$$

(b) Time response =

$$\boxed{L^{-1}\left[sI-A\right]^{-1} x(0) = \phi(t)\, x(0)}$$

2×2 $\quad$ 2×1 $\qquad$ $x(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$.

$$\Rightarrow \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{4t} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ e^{4t} \end{bmatrix}$$

(c) solution to state equation if input is unit impulse

$$x(t) = L^{-1}[SI - A]^{-1}x(0) + L^{-1}[(SI - A)]^{-1}BU(s)$$

$$= \begin{bmatrix} 0 \\ e^{4t} \end{bmatrix} + L^{-1}\left[(SI - A)^{-1}BU(s)\right]$$

$u(s) = \text{unit} = 1$

$$\begin{bmatrix} 0 \\ e^{4t} \end{bmatrix} + L^{-1}\left[\begin{pmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-4} \end{pmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}\begin{bmatrix} 1 \end{bmatrix}\right]$$

2×2 2×1

2×1

$$= \begin{bmatrix} 0 \\ e^{4t} \end{bmatrix} + L^{-1}\begin{bmatrix} \frac{1}{s-2} \\ \frac{1}{s-4} \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ e^{4t} \end{bmatrix} + \begin{bmatrix} e^{2t} \\ e^{4t} \end{bmatrix} = \begin{bmatrix} e^{2t} \\ 2e^{4t} \end{bmatrix}$$

(d) Output

$$y = [4 \ \ 0] \, x(t)$$

$$= [4 \ \ 0] \begin{bmatrix} e^{2t} \\ 2e^{4t} \end{bmatrix} = \underline{\underline{4e^{2t}}}.$$

(Q) Find state transition matrix and time response.

$$\dot{x} = \overset{A}{\begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \qquad x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

(a) State transition matrix.

$$\phi(t) = L^{-1}\left[sI - A\right]^{-1}$$

$$[sI - A] = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 \\ 2 & s \end{bmatrix}.$$

$$[s\underline{I}-A]^{-1} = \frac{\begin{bmatrix} s & 1 \\ -2 & s \end{bmatrix}}{s^2+2}$$

$$L^{-1}\left[s\underline{I}-A\right]^{-1}$$

$$= L^{-1}\begin{bmatrix} \frac{s}{s^2+2} & \frac{1}{s^2+2} \\ \frac{-2}{s^2+2} & \frac{s}{s^2+2} \end{bmatrix}$$

$$= \begin{bmatrix} \cos\sqrt{2}t & \frac{\sin\sqrt{2}t}{\sqrt{2}} \\ \frac{-2\sin\sqrt{2}t}{\sqrt{2}} & \cos\sqrt{2}t \end{bmatrix}$$

(b) time response =>

$$L^{-1}\left[s\underline{I}-A\right]^{-1} \underset{2\times 2 \quad 2\times 1}{x(0)} = \phi(t)\; x(0)_{2\times 1}$$

$$= \begin{bmatrix} \cos\sqrt{2}t & \frac{\sin\sqrt{2}t}{\sqrt{2}} \\ \frac{-2\sin\sqrt{2}t}{\sqrt{2}} & \cos\sqrt{2}t \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

$$= \begin{bmatrix} \cos\sqrt{2}t + \frac{\sin\sqrt{2}t}{\sqrt{2}} \\ \frac{-2\sin\sqrt{2}t}{\sqrt{2}} + \cos\sqrt{2}t \end{bmatrix} //$$

(Q) Obtain the unit step response of the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

for $(x)(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Ans) $x(t) = \underbrace{L^{-1}\left[sI - A\right]^{-1} x(0)}_{=0 \text{ since } x(0)=0} + L^{-1}\left[(sI-A)\right]^{-1} B\,u(s)$

$\therefore L^{-1}\left[(sI-A)^{-1} B\,u(s)\right]$

$= L^{-1}$

$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

$= \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$

$(sI-A)^{-1} = \dfrac{\begin{bmatrix} 3+s & 1 \\ -2 & s \end{bmatrix}}{s(s+3)+2}$

$$= \frac{\begin{bmatrix} 3+s & 1 \\ -2 & s \end{bmatrix}}{s^2+3s+2} \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} \begin{bmatrix} 1/s \end{bmatrix}$$

$$\begin{array}{cc} 2\times 2 & 2\times 1 \end{array}$$

$$= L^{-1}\left[\begin{bmatrix} \frac{3+s}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}\begin{bmatrix} \frac{1}{s} \end{bmatrix}\right]$$

$$= L^{-1}\left[\begin{bmatrix} \frac{1}{(s+1)(s+2)} \\ \frac{s}{(s+1)(s+2)} \end{bmatrix}\begin{bmatrix} \frac{1}{s} \end{bmatrix}\right]$$

$$= L^{-1}\begin{bmatrix} \frac{1}{s(s+1)(s+2)} \\ \frac{1}{(s+1)(s+2)} \end{bmatrix}$$

$$\frac{1}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$1 = A(s+2) + B(s+1)$$

put $s = -2$ $\quad 1 = B(-1)$ $\quad \underline{B = -1}$ $\quad$ and $A = 1$

$$\frac{1}{(s+1)(s+2)} = \frac{1}{(s+1)} - \frac{1}{(s+2)}.$$

$$\cancel{\frac{1}{s(s+1)(s+2)} = \frac{1}{(s+1)} - \frac{1}{s+1}}$$

$$\frac{1}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}.$$

$$A = 1/2 \quad B = -1 \quad C = 1/2.$$

$$\frac{1}{s(s+1)(s+2)} = \frac{\frac{1}{2}}{s} - \frac{1}{s+1} + \frac{\frac{1}{2}}{s+2}.$$

$$x(t) = L^{-1}\begin{bmatrix} \frac{1/2}{s} - \frac{1}{s+1} + \frac{1/2}{s+2} \\ \frac{1}{s+1} - \frac{1}{s+2} \end{bmatrix}.$$

$$= \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2} e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}.$$

(Q) Find the state transition matrix using cayley Hamilton theorem is

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Ans) : $\boxed{\left[\lambda I - A\right] = 0}$

$$\begin{vmatrix} \lambda & -1 \\ 2 & \lambda+3 \end{vmatrix} = 0$$

$$\lambda(\lambda+3) - (-1 \times 2)$$

$$= \lambda^2 + 3\lambda + 3 = 0$$

$$= (\lambda+1)(\lambda+2) = 0$$

$$\lambda = -1 \text{ or } \lambda = -2$$

$$\boxed{R(\lambda) = a_0 + a_1 \lambda = e^{\lambda t}}$$

$$R(-1) = e^{-t} = a_0 + a_1(-1)$$

$$R(-2) = e^{-2t} = a_0 + a_1(-2).$$

$$a_0 - a_1 = e^{-t}$$

$$a_0 - 2a_1 = e^{-2t}$$

$$0 \quad a_1 = e^{-t} - e^{-2t}$$

$$a_0 = 2e^{-t} - e^{-2t}$$

STM $\boxed{e^{At} = a_0 \underline{I} + a_1 A}$

$$\overset{1\times 2}{2e^{-t} - e^{-2t}} \overset{2\times 2}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} +$$

$$(e^{t} - e^{-2t}) \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 2e^{-t} - e^{-2t} & 0 \\ 0 & 2e^{-t} - e^{-2t} \end{bmatrix} +$$

$$\begin{bmatrix} 0 & e^{t} - e^{-2t} \\ -2\left[(e^{t} - e^{-2t})\right] & -3\left(2e^{-t} - e^{-2t}\right) \end{bmatrix}$$

$$\phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{t} - e^{-2t} \\ -2e^{t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

Discrete time Systems.

$$x(k+1) = A\,x(k) + B\,u(k)$$
$$y(k) = C\,x(k).$$

(Q) Find T.F if

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

A)

$$T.F = C\left[zI - A\right]^{-1} B + D$$

$$\left[zI - A\right] = \begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} z & -1 \\ 6 & z+5 \end{bmatrix}$$

$$\left[zI - A\right]^{-1} = \frac{\begin{bmatrix} z+5 & 1 \\ -6 & z \end{bmatrix}}{z^2 + 5z + 6}$$

$$C[zI-A]^{-1}B = \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{\begin{bmatrix} z+5 & 1 \\ -6 & z \end{bmatrix}}{z^2+5z+6} \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

1×2 2×1

$$= \frac{\begin{bmatrix} z+5 & 1 \end{bmatrix}}{z^2+5z+6} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{z+5}{z^2+5z+6}$$

(Q) Find the state transition matrix of

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(k).$$

$$\text{STM } \phi k = Z^{-1}\left[\left[(zI - A) \right]^{-1} z \right].$$

$$(zI - A) = \begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} z+3 & 0 \\ 0 & z+2 \end{bmatrix}.$$

$$(zI - A)^{-1} = \frac{\begin{bmatrix} z+2 & 0 \\ 0 & z+3 \end{bmatrix}}{(z+3)(z+2)}$$

$$= \begin{bmatrix} \frac{1}{z+3} & 0 \\ 0 & \frac{1}{z+2} \end{bmatrix}$$

$$z^{-1}\left\{\left[zI - A\right]^{-1} z\right\} = \begin{bmatrix} \frac{z}{z+3} & 0 \\ 0 & \frac{z}{z+2} \end{bmatrix}$$

$$\boxed{z^{-1}\left[\frac{z}{z+a}\right] = (-a)^{k}}$$

$$\phi(k) = \begin{bmatrix} (-3)^{k} u(k) & 0 \\ 0 & (-2)^{k} u(k) \end{bmatrix}$$

(Q) A discrete time systems is described by difference equation

$$y(k+2)+5y(k+1)+6y(k)=u(k)$$

$$y(0)=y(1)=0 \qquad T=1\,sec.$$

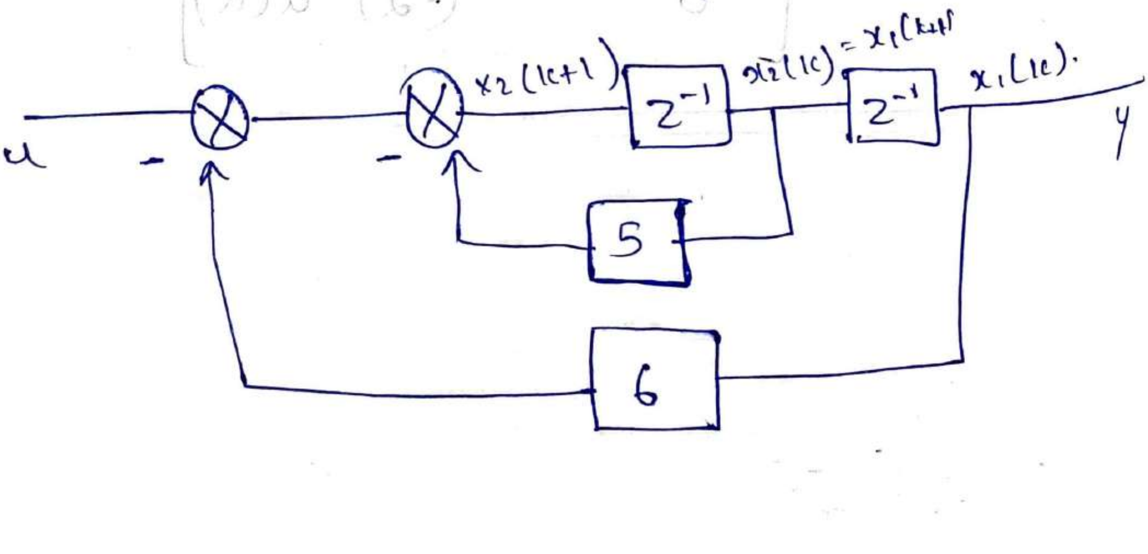
Find state model in controllable canonical form

Ans)

$$\cancel{z^2(y(z)+5z}$$

$$z^2 Y(z)+5zY(z)+6Y(z)=U(z)$$

$$\frac{Y(z)}{U(z)}=\frac{1}{z^2+5z+6}$$



$x_1(k+1) = x_2(k)$

$x_2(k+1) = -6x_1 - 5x_2 + u$.

$y = x_1(k)$.

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k).$$

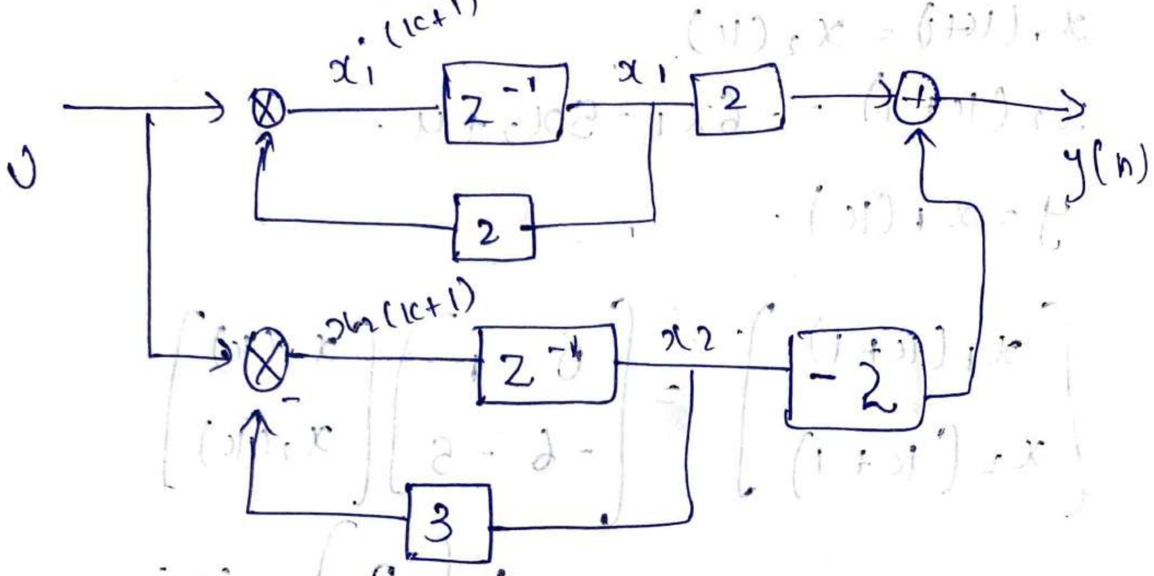
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$$

(Q) Find the canonical or diagonal form or parallel form of $\frac{Y(z)}{X(z)} = \frac{2}{(z+2)(z+3)}$.

Ans) $\frac{2}{(z+2)(z+3)} = \frac{A}{z+2} + \frac{B}{z+3}$.

$A = 2 \quad B = -2$.

$= \frac{2}{z+2} - \frac{2}{z+3}$.



(Q) obtain the unit step response, $y(t)$ of the system

$$\dot{x} = \begin{bmatrix} -1 & 0 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u.$$

$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} x.$$

A) $x(t) = L^{-1}\left[SI - A\right]^{-1} x(0) +$

$\quad \overset{\shortparallel}{0}$

$$L^{-1}\left[(SI - A)^{-1} B\, u(s)\right].$$

$u(s) = 1/s.$

$$L^{-1}\left[(sI-A)\right]^{-1} \cdot B\,U(s)$$

$$sI-A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} s+1 & 0 \\ 2 & s+3 \end{bmatrix}$$

$$(sI-A)^{-1} = \frac{\begin{bmatrix} s+3 & 0 \\ -2 & s+1 \end{bmatrix}}{\left[(s+3)(s+1)\right] - [0]}.$$

$$= \begin{bmatrix} \frac{1}{s+1} & 0 \\ \frac{-2}{(s+3)(s+1)} & \frac{1}{(s+3)} \end{bmatrix}.$$

2×1

$$(sI-A)^{-1} \cdot B \cdot U(s) = \begin{bmatrix} \frac{1}{s+1} & 0 \\ \frac{-2}{(s+3)(s+1)} & \frac{1}{s+3} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{1}{s+1} \\[2ex] \dfrac{2}{(s+3)(s+1)} + \dfrac{1}{(s+3)} \end{bmatrix} \dfrac{1}{s}$$

$$= \begin{bmatrix} \dfrac{1}{(s+1)s} \\[2ex] \dfrac{-2}{s(s+3)(s+1)} + \dfrac{1}{s(s+3)} \end{bmatrix}$$

Do partial fraction and find L^{-1}.

$$\frac{A}{s} + \frac{B}{s+1}$$

$$\frac{A}{s} + \frac{B}{s+3} + \frac{C}{s+1} \qquad \frac{A}{s} + \frac{B}{s+1}$$

(Q) $X = \begin{bmatrix} -1 & 1 \\ 1 & -2 \end{bmatrix} X$.

a) $\phi(t) = L^{-1}[SI - A]^{-1}$,

$$SI - A = \begin{bmatrix} S & 0 \\ 0 & S \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} S+1 & -1 \\ -1 & S+2 \end{bmatrix}.$$

$$(SI - A)^{-1} = \frac{\begin{bmatrix} S+2 & +1 \\ 1 & S+1 \end{bmatrix}}{(S+1)(S+2) - (+1 \times 1)}.$$

$$= \frac{\begin{bmatrix} S+2 & -1 \\ 1 & S+1 \end{bmatrix}}{(S+1)(S+2) - 1}$$

$\underline{S^2 + 2S + S + 2 - 1}$ $\quad S^2 + 2S + S + 2 - 1$

$=$ $\quad = S^2 + 3S + 1$

$$(SI - A)^{-1} = \begin{bmatrix} \dfrac{s+2}{s^2+3s+1} & \dfrac{-1}{s^2+3s+1} \\ \dfrac{1}{s^2+3s+1} & \dfrac{s+1}{s^2+3s+1} \end{bmatrix}$$

$$\begin{bmatrix} \dfrac{s+2}{\left(s-\left(\frac{-3-\sqrt{5}}{2}\right)\right)\left(s-\frac{-3+\sqrt{5}}{2}\right)} & \dfrac{-1}{(\quad)(\quad)} \\ \dfrac{1}{(\quad)(\quad)} & \dfrac{s+1}{(\quad)(\quad)} \end{bmatrix}$$

Time response of state space.

$$x^{\cdot}(t) = A\,x(t) + Bu(t)$$

taking L·T

$$sX(s) - x(0) = A\,X(s) + BU(s)$$

$$X(s)[sI - A] = x(0) + Bu(s)$$

$$X(s) = [sI - A]^{-1}x(0) + [sI - A]^{-1}Bu(s)$$

$$x(t) = \underbrace{L^{-1}[sI - A]^{-1}x(0)}_{\text{zero i/p resp}} + \underbrace{L^{-1}[sI - A]^{-1}BU(s)}_{\text{zero o/p resp.}}$$

where $L^{-1}[sI - A]^{-1} = \phi(t)$ is called state transition matrix.

Properties of state transition matrix

$\dot{x} = Ax$ is $u = 0$

$\phi(t) = e^{At}$

(a) $\phi(0) = I$.

(b) $\phi^{-1}(t) = [e^{At}]^{-1} = e^{-At} = \phi(-t)$

$$\boxed{\phi^{-1}(t) = \phi(-t)}$$

(c) $[\phi(t)]^{k} = [e^{At}]^{k} = e^{Akt} = \phi(kt)$.

$\therefore \boxed{[(\phi(t))^{k}] = \phi[kt]}$

(d) $\boxed{\dot{\phi}(t) = A\phi(t)}$

$\dot{\phi}(t) = \frac{d}{dt} e^{At} = Ae^{At} = A\phi(t)$

(e) $\boxed{\phi(t_2 - t_1)\,\phi(t_1 - t_0) = \phi(t_2 - t_0)}$

$= e^{A[t_2 - t_1]}\, e^{A[t_1 - t_0]}$

$= e^{At_2}\, e^{-At_1}\, e^{At_1} e^{-At_0}$

$= e^{A(t_2 - t_0)} = \phi(t_2 - t_0)$.

Transfer Function or transfer matrix:

$\dot{x} = Ax + Bu \longrightarrow$ state eqn

$sX(s) - X(0) = AX(s) + BU(s)$. taking L.T

$x(0) = 0$

$sX(s) = AX(s) + BU(s)$.

$X(s) = [sI - A]^{-1} BU(s)$

$y = Cx + Du \longrightarrow$ o/p eqn.

$Y(s) = CX(s) + DU(s)$.

$Y(s) = C[sI - A]^{-1} BU(s) + DU(s)$

$$T.F = \frac{Y(s)}{U(s)} = C[sI - A]^{-1}B + D$$

For discrete time

$$T.F = \frac{Y(z)}{U(z)} = C[zI - A]^{-1}B + D$$